General Instruction

1. Student Instruction

Basic Rules

1. Each team must consist of 1–4 students.
2. Every team must submit via GitHub Repository Link
    - Working Project (Frontend + Backend)
    - Final Presentation (PPT and Demo)
3. Project must be:
    - Original
    - Developed during the hackathon timeline
4. AI tools (ChatGPT, Copilot, etc.) are allowed for guidance, but:
    - You must understand your code.
    - You must explain your logic during judging.

Timeline Discipline

- No late submissions.
- Final Git commit time will be verified.
- Last-minute ZIP submissions without proper repository history will be rejected.

Project Expectations

Your project must include:

- Working UI
- Proper Backend/API
- Database integration
- Deployment (recommended and Preferable)

2. Git Repository & Team Development Rules

Mandatory Requirements

1. Repository must be created by the team leader within 24 hours of releasing the problem statement.
2. Repository name format: NavKalpana-TeamCode
   Example: NavKalpana-RICR-NK-0001
3. All team members must:
    - Must be the contributor in repo
    - Use individual GitHub accounts
    - Contribute via commits
    - Avoid single-person commits for team projects

Commit Rules

- Minimum 10 meaningful commits.
- Avoid a single bulk commit at the end.
- Commit messages must be clear.
    - Good Commit Example:
        - Added login authentication
        - Integrated payment API
        - Fixed cart validation bug

- Bad Commit Example:
  - Update
  - Final
  - Added
  - done

Bad Commit Messages will lead to negative marking

Branching Rule (Recommended)

- main – Final stable version
- All the Members must Create Separate Branches which further merges to the Main Branch
  - Example:
    - feature/login
    - feature/register
    - bug/update etc.

3. Recommended Project Folder Structure

Repository Name Example: *NavKalpana-RICR-NK-0001*

Standard Structure

- NavKalpana-TeamCode
  - frontend
  - backend
  - docs
    - problem-statement.pdf
    - architecture-diagram.png
    - api-documentation.md
    - presentation.pptx
  - README.md

4. README.md Must Include

- Project Title
- Team Members & Roles
- Problem Statement
- Tech Stack Used
- Installation Steps
- API Endpoints
- Screenshots
- Future Improvements

5. What Judges Will Verify

Technical Verification

- Git commit history
- Contribution by all members
- Code understanding

- Folder structure clarity
- Backend API functionality
- Database integration
- Error handling

Innovation Check

Judges will evaluate:

- Whether it solves a real problem
- Practical usability
- Scalability potential
- Meaningful use of AI (if included)

Viva Round Questions

Each member may be asked:

- Explain your module.
- Explain database schema.
- Explain API flow.
- Explain deployment steps.
- Explain one technical challenge you faced.

6. Important Warning

Immediate disqualification may occur if:

- Code is copied from the internet without modification
- Purchased templates are used without proper logic implementation
- Repository is created only on the final day
- Only UI is presented without functionality
- Fake data is shown without backend logic (unless clearly defined prototype)
- Only one member answers all questions

7. Bonus Considerations

- Clean UI/UX
- Proper Authentication
- Role-based access control
- AI/ML Integration (if required)
- Deployment (Vercel, Render, AWS, Azure)
- Clean code practices

Final Note

The hackathon is not only about coding. It is about teamwork, structured thinking, problem solving, and clear communication. Build a meaningful solution and be prepared to explain it confidently.

Academic Operations and Management Portal
(Teacher Module)

Product Overview
The Academic Operations and Management Portal is a secure web-based system designed for authorized faculty members to:

- Manage academic content
- Monitor student performance
- Evaluate assignments and quizzes
- Manage attendance
- Analyze learning outcomes

The system is divided into:

- **Part 1 – Core Academic Management**
- **Part 2 – Analytics, Monitoring & Growth Intelligence**

Global Layout & Navigation
Sidebar Navigation (Mandatory)
The Portal must include a persistent left sidebar with:

- Dashboard
- Batch Management
- Assessment Management
- Student Management
- Support Requests

The active section must be visually highlighted.
Top Navigation Bar (Mandatory)
The top bar must include:

- Time-based greeting with Teacher Name
- Teacher Profile Section (Profile Image, Name, Designation)
- Profile Dropdown
- Logout Option

Example:
Good Morning, [Teacher Name]

Part 1 – Core Academic Management

1. Authentication Module
Objective
Provide secure access to authorized teachers only.
Features

- Email and Password login
- Role-based access control (Teacher role)
- Password hashing (bcrypt recommended)
- JWT-based session management
- Logout functionality
- Session timeout (approximately 24 hours)

Functional Flow
Login → Validate Credentials → Generate JWT → Redirect to Teacher Dashboard

2. Dashboard
The Teacher Dashboard is the landing page after login.
KPI Summary Cards (Mandatory)
The dashboard must display:
- Total Students
- Active Courses
- Pending Assignments for Evaluation

Dynamic Update Conditions
These cards must dynamically update when:
- Students are enrolled or removed
- Courses are created or archived
- Assignments are evaluated

Cards must be color-coded for clarity.
Additional Components
- Upcoming Deadlines
- Average Class Performance
- Quick Navigation Buttons
- Optional Weekly Activity Chart
- Optional Low-Performing Student Alerts

3. Batch Management
This module allows faculty to manage, monitor, and control batches, and student attendance.
There Should be Dummy Course

3.1 Batch Management (Card-Based Layout)
Each batch card must display:
- Batch Name
- Batch Type (e.g., WEB Batch, DSA Batch)
- Total Students Enrolled
- Progress Percentage
- Status Tag (Ongoing / Completed / Upcoming)
- Manage Button (This opens the Attendance Management)
- End Button

Progress must be shown using a visual progress bar.
3.2 Batch Filtering
The Modules page must include filter tabs:
- Ongoing
- Completed
- Upcoming

Filters must dynamically update visible batch cards.

3.3 Attendance Management

Teachers must be able to:

- Mark attendance per session
- Add Remarks before Submission (Mandatory)
- View attendance history
- Edit attendance records (Same Date Only & Active till 10 mins after Submission)
- Export attendance records

Enhanced Attendance View

Attendance must support:

- Calendar view per student
- Present / Absent / Late classification
- Module-wise attendance breakdown
- Date-wise remarks

Attendance records must be visible in:

- Batch context
- Individual student profile

Synchronization Requirements

Attendance updates must reflect in:

- Student Attendance Section (Under Student Management Section)
- Analytics metrics
- Overall Growth Index calculations

4. Assessment Management

This module defines the academic hierarchy:

All academic activities are managed at the lesson level.

4.1 Lesson-wise Assignment Management

Assignment Creation

Teachers must be able to:

- Create assignments under a lesson
- Edit assignments
- Set deadline
- Attach instructions
- Define submission type (Image,PDF,JPG etc)

Each assignment must include:

- Title
- Description
- Deadline
- Maximum Marks

Assignment Evaluation

Teachers must be able to:

- View submissions
- Download files
- Detect late submissions
- Enter marks
- Provide feedback

- Mark as Evaluated

Evaluation must update:

- Student Progress Tracker (Under Student Management Section)
- OGI calculations

Assignment status must reflect:

- Not Submitted
- Submitted
- Late Submitted
- Evaluated

4.2 Lesson-wise Quiz Management

Quiz Creation

Teachers must be able to:

- Create quizzes under lessons
- Set duration
- Add MCQ questions
- Define answer types
- Add explanations
- Set attempt limits

Each quiz must include:

- Title
- Duration
- Question list
- Total marks

Quiz Monitoring

Teachers must be able to:

- View attempts
- View scores
- View class average
- Restrict attempts

Quiz updates must reflect in:

- Student Progress Tracker (Under Student Management Section)
- OGI calculations

5. Student Management

Students List View

Teachers must be able to:

- Search by Name, Enrollment ID, Email, or Course
- Filter by Course

Summary Cards:

- Total Students
- Ongoing Students
- Completed Students

Students grouped into:

- Ongoing
- Completed

Student Table View

Each row must display:

- Avatar
- Student Name
- Enrollment ID
- Course
- Enrolled Modules (tags)
- Attendance Percentage
- Email
- Phone
- GitHub & LinkedIn

Attendance color coding:

- Green (High)
- Yellow (Medium)
- Red (Low)

Clicking opens a detailed modal.

Student Detail Modal

Header:

- Student Name
- Course
- Enrollment ID
- Status Badge
- Attendance Badge

Tabs:

1. Course Information
2. Progress
3. Attendance

Course Information Tab

- Current Course
- Enrollment Number
- Weekly Activity Chart
- Modules (Completed / Total)
- Skills Acquired
- Learning Streak
- Attendance Summary

Progress Tab

- Module-wise completion
- Assignment summary
- Quiz summary
- Overall Progress Percentage

Attendance Tab

- Overall Attendance Percentage
- Total Present
- Total Absent
- Total Late
- Calendar View
- Module-wise stats
- Date-wise remarks

Note: You can use Dummy Data Wherever Required.

6. Support Requests

Support page must include:

- Course filter
- Status filter (Pending / Resolved)
- Student name
- Topic
- Description
- Attachment preview

Teachers must be able to:

- View doubts
- Filter by course and topic
- Reply to doubts
- Upload solution files
- Mark as resolved

Teachers must also approve or schedule backup classes.